

MOON PHASES EXPLORER



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Chapter 1: Introduction

1.1. Abstract

Moon Phases Explorer is an interactive computer graphics project designed to visually simulate the movement of the Moon around the Earth and demonstrate the main phases of the Moon. The project uses OpenGL and GLUT to create a 3D space environment that includes the Earth, the Moon, orbit path, stars, lighting effects, and phase markers. The main goal of the project is to help users understand the relationship between the Moon's position and its visible phases in an interactive and visual way. The contribution of this project is combining educational content with graphical interaction, allowing the user to select moon phases, pause the animation, zoom in and out, and rotate the camera to explore the scene from different angles.

1.2. Introduction

The Moon is one of the most important objects observed from Earth. Its appearance changes during the month because of its position relative to the Earth and the light source. These changes are known as moon phases. Understanding moon phases can be difficult through text only, so this project presents the concept visually using a 3D interactive simulation.

Moon Phases Explorer is a graphical application that displays the Earth at the center and the Moon moving around it. The system shows the Moon's orbit and marks the eight main moon phases: New Moon, Waxing Crescent, First Quarter, Waxing Gibbous, Full Moon, Waning Gibbous, Last Quarter, and Waning Crescent. Each phase has a specific position on the orbit and includes a short description.

The project allows the user to interact with the simulation using the mouse and keyboard. By clicking on a moon phase marker, the selected phase is highlighted, the Moon is enlarged, and information about the phase is displayed on the screen. The user can also pause or resume the animation, rotate the camera, zoom in and out, and reset the view. This makes the project more interactive and easier to understand.

The scope of the project focuses on educational visualization rather than scientific accuracy. The project aims to present the general idea of moon phases using simple 3D graphics, animation, lighting, textures, and user interaction.

Chapter 2: Methodology

2.1. Functionality and Implementation Details

The project was implemented using C++ with OpenGL, GLU, GLUT, and GLEW. These libraries were used to create the 3D scene, render objects, manage textures, handle lighting, and receive user input from the keyboard and mouse.

The main functionalities of the project are:

1. **3D Space Environment**

The project creates a space-like environment with a dark background and many stars. The stars are generated using arrays that store their positions and brightness values. This gives the scene a realistic space appearance and makes the simulation visually attractive.

2. **Earth Rendering**

The Earth is drawn as a 3D sphere using gluSphere. The program attempts to load an Earth texture from the assets folder. If the texture is not found, the system creates a fallback texture automatically. The Earth also rotates around itself to make the scene more dynamic.

3. **Moon Rendering and Orbit**

The Moon is also drawn as a textured 3D sphere. It moves around the Earth using an orbit angle and a fixed orbit radius. The Moon's position is calculated using trigonometric functions such as cos and sin. The project also draws a circular orbit line to show the path of the Moon around the Earth.

4. **Moon Phase Markers**

The system includes eight phase markers placed around the Moon's orbit. Each marker represents one of the main moon phases. These markers help the user identify the phase positions. When a phase is selected, its marker is highlighted with a different color.

The eight phases used in the project are:

- New Moon
- Waxing Crescent

- First Quarter
- Waxing Gibbous
- Full Moon
- Waning Gibbous
- Last Quarter
- Waning Crescent

5. **User Interaction**

The project supports several controls:

- Left mouse click: Select a moon phase.
- Space key: Pause or resume the animation.
- Arrow keys: Rotate the camera.
-
    - + / - keys: Zoom in and zoom out.
- R key: Reset the camera and animation.
- ESC key: Exit the application.

These controls allow the user to explore the simulation freely and understand the moon phases more clearly.

6. **Phase Information Display**

When the user selects a moon phase, the project displays the phase name and a short explanation on the screen. A small 2D moon phase preview is also shown in the corner of the screen. This preview gives the user a clearer visual idea of the selected phase.

7. **Lighting**

The project uses OpenGL lighting to illuminate the Earth and Moon. A light source is placed in the scene, although the light object itself is not drawn. This creates visible bright and dark areas on the objects, which supports the idea of moon phases.

8. **Texture Loading**

The project includes a texture loading function that reads BMP image files. It loads Earth and Moon textures from the assets folder. If the images are missing, the program creates simple fallback textures so the project can still run without errors.

2.2. Additional Libraries Used

The following libraries were used in the project:

- **OpenGL:** Used for rendering 2D and 3D graphics.
- **GLU:** Used for drawing spheres and setting the perspective projection.
- **GLUT / FreeGLUT:** Used for window creation, keyboard input, mouse input, and the main loop.
- **GLEW:** Used to initialize OpenGL extensions on non-Apple systems.
- **cmath:** Used for mathematical calculations such as sine, cosine, and square root.
- **cstdio /cstdlib /iostream:** Used for file handling, console output, and program control.

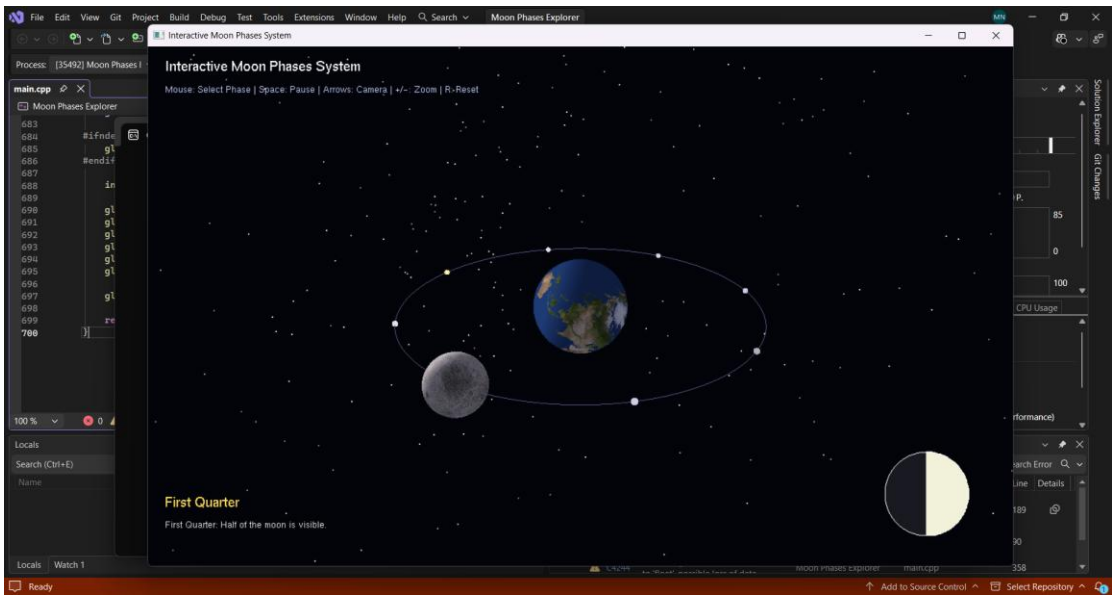
Chapter 3: Result

3.1. Screenshots with Description

This chapter presents the final output of the Moon Phases Explorer project.

Figure 1: Main Simulation Window

This screenshot shows the main 3D scene of the project. The Earth is displayed at the center, and the Moon moves around it in a circular orbit. The background contains stars to represent outer space. The orbit line and phase markers are also visible around the Earth.

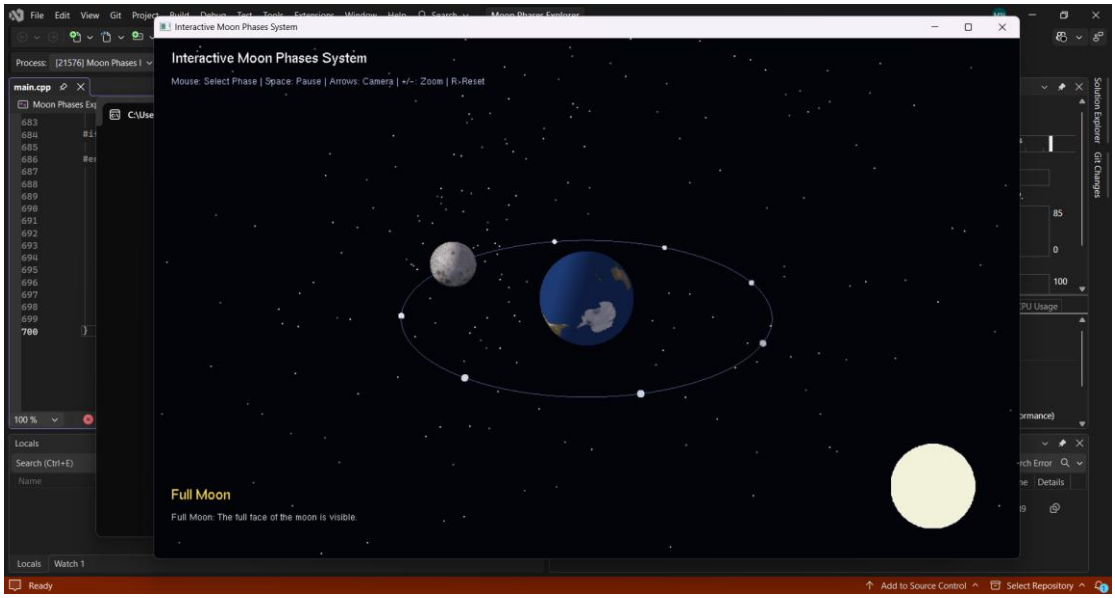


Description:

The main window demonstrates the general structure of the project. It shows the Earth, the Moon, the orbit path, and the space environment. The camera view allows the user to observe the system from a clear angle.

Figure 2: Selected Moon Phase Information

This screenshot shows what happens when the user clicks on a moon phase marker. The Moon becomes enlarged, the animation pauses, and the selected phase name and description appear on the screen.

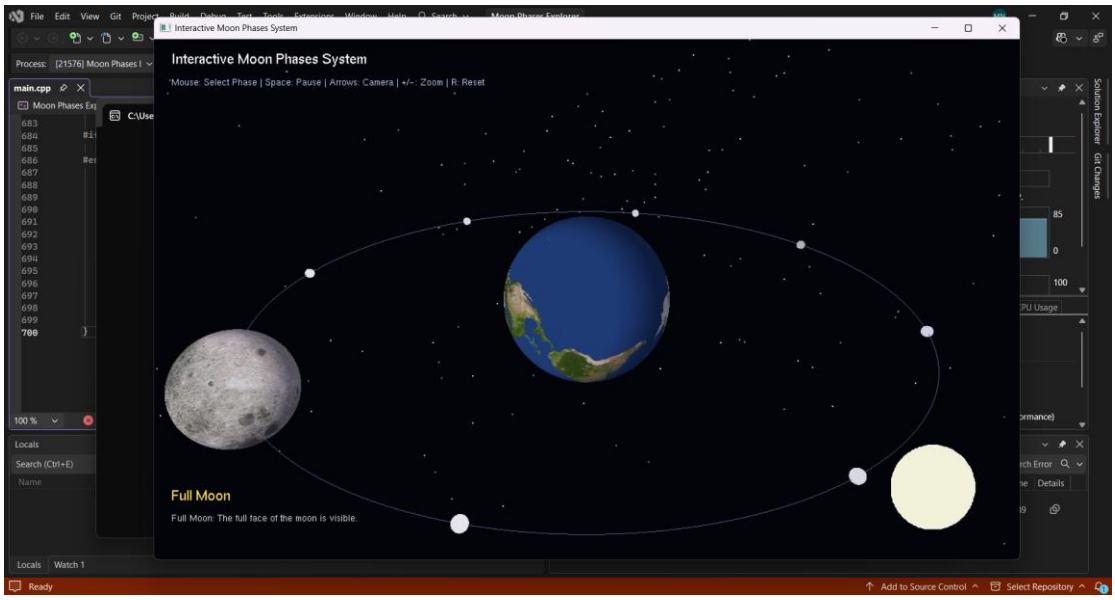


Description:

This feature improves the learning experience by giving the user direct information about the selected phase. The small moon phase preview also helps the user understand how the Moon appears during that phase.

Figure 3: Camera Rotation and Zoom

This screenshot shows the scene after changing the camera angle and zoom level using the keyboard controls.



Description:

The camera controls allow the user to explore the scene from different views. This makes the simulation more flexible and helps the user understand the 3D arrangement of the Earth, Moon, and orbit.

Conclusion and Future Work

Moon Phases Explorer successfully presents an interactive 3D simulation of the Moon moving around the Earth. The project combines graphics, animation, lighting, textures, and user interaction to explain the main moon phases in a simple and visual way. The user can select phases, read short descriptions, pause the animation, rotate the camera, and zoom in or out.

The project achieved its main goal by creating an educational and interactive environment for exploring moon phases. It also helped demonstrate important computer graphics concepts such as 3D transformations, texture mapping, lighting, camera control, and event handling.

In the future, the project could be improved by adding more realistic astronomical calculations, a visible Sun object, smoother textures, sound effects, labels for each phase, and a timeline that shows the days of the lunar cycle. Another possible improvement is adding a graphical user interface menu that allows users to choose the speed of the animation or switch between educational and realistic modes.

References

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